

ME 626: Acoustics

Course Code	ME 626
Course Name	Acoustics
L-T-P-C	3-0-0-3
Prerequisites	Basic Engineering Mathematics. ME 602 Mechanical Vibration is desirable
Intended for	MS and PhD students – SE, SCEE, B-Tech final and pre-final year students.
Distribution	Elective
Approval	9th Senate (May 2016)

Course Contents

- **Basics of Acoustics and Vibration**
Introduction, frequency, Fourier series, FFT, loudness, decibel scale, octave, music scale. Vibration of one degree of freedom system. Brief introduction to multi degree of freedom system. [5 Lectures]
- **Vibration of string, bars, membranes and plates**
One dimensional wave equation, general solution of wave equation, plucked string. Longitudinal vibration of bars, transverse vibration of beam. Wave equation for stretched membrane, normal modes of membrane. Vibration of plates. [5 Lectures]
- **Acoustic wave equation and its solution**
Equation of state, continuity, Euler's equation. Linear wave equation, speed of sound in fluids. Acoustic intensity, specific acoustic impedance, spherical waves, cylindrical waves. Waveguides, transmission line equations. and resonators. [8 Lectures]
- **Reflection and transmission**
Changes in media, transmission from one fluid to another – normal and oblique incidence. Normal specific acoustic impedance, reflection from solid. Transmission through thin partition – mass law. Perfectly matched layer. [4 Lectures]
- **Radiation and reception of acoustic waves**
Radiation from pulsating sphere, acoustic reciprocity, continuous line source, radiation from circular piston, radiation impedance. Models of electromechanical acoustic systems, Solution for a loudspeaker model, Microphones. [8 Lectures]
- **Hearing and Speech**
The ear, fundamental properties of hearing – threshold, equal loudness level contours, critical bandwidth, masking, beats. Loudness level and loudness, pitch and frequency, voice. [4 Lectures]
- **Environmental and Architectural acoustics**
Sound in enclosure, reverberation time, sound absorption materials, acoustics factors in architectural design. Speech interference, noise rating curves, highway noise, design of partitions. [8 Lectures]
- **Special topics**
Underwater acoustics, shock waves and explosion. (can be in the form of seminars/presentation) [2 Lectures]

Text Books

1. Lawrence E. Kinsler, Austin R. Frey, **Fundamentals of Acoustics**, 4th Edition, Wiley Publishers, 2000.
2. Munjal M. L., **Noise and Vibration Control**, World Scientific Publishers in Collaboration with IISc Press, Singapore, 2013

References

1. Beranek Leo L., **Acoustics**, Acoustical Society of America, 1996.